



B.Tech. Degree VI Semester Examination April 2014

ME 604 HEAT AND MASS TRANSFER (2006 Scheme)

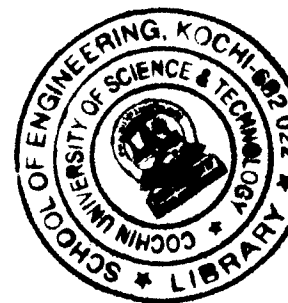
Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Derive an expression for heat flow rate and temperature distribution for an infinite hollow cylinder.
- (b) Define and explain the physical significance of thermal conductivity and thermal diffusivity.
- (c) Explain the phenomenon of film wise and drop wise condensation with neat diagrams.
- (d) Differentiate between diffusive and convective mass transfer.
- (e) State and explain Kirchoffs' identity.
- (f) Derive an expression for radiation heat flux between two parallel infinite surfaces.
- (g) How are heat exchangers classified? Sketch temperature variation in parallel and counter flow heat exchangers.
- (h) What design aspects are to be considered while designing heat exchangers?



PART B

(4 x 15 = 60)

- II. (a) Derive the differential equation of heat conduction in Cartesian co-ordinate system. (10)
 - (b) A steel ball 5cm in diameter and initially at a temperature of 450°C is suddenly placed in a controlled environment where the temperature is maintained at 100°C. Calculate the time required for the ball to attain a temperature of 150°C. Take the following property values: (5)
- $K = 35 \text{ W/mK}$, $C_p = 0.46 \text{ KJ/KgK}$
 $S = 7800 \text{ kg/m}^3$ and $h = 10 \text{ W/m}^2 \text{ K}$

OR

- III. (a) An exterior wall of house may be approximated by a 100mm layer of common brick ($K = 0.7 \text{ W/mK}$) followed by a 40mm layer of gypsum plaster ($K = 0.48 \text{ W/mK}$). What thickness of loosely packed rock wool insulation ($K = 0.065 \text{ W/mK}$) should be added to reduce the heat loss through the wall by 75%? Assume the inside air at 20°C while ambient air at 35 °C. Take heat transfer coefficient on both sides as 15 W/m²K. (9)
- (b) The tip of a soldering gun consists of 6mm diameter Cu rod with $K = 380 \text{ W/mK}$ and 75mm long. If the tip must be at 200 °C what should be the temperature at base and the heat transfer from the rod. Assume surrounding temperature = 20 °C and $h = 25 \text{ W/m}^2 \text{ K}$ (6)

(P.T.O.)

- IV. Air at 27°C and 1 bar flows over a flat plate at 2m/s . The plate is maintained at 60°C . Calculate the heat transfer from the portion between 20cm and 40cm from the leading edge of plate. Assume unit width. (15)

OR

- V. (a) Define and give physical significance of the following: (6)
- (i) Nusselt's Number
 - (ii) Grashof's Number
 - (iii) Prandtl's Number
- (b) Find rate of heat loss from cubical furnace kept on a concrete floor if the outside temperature of furnace is 80°C and surrounding air is at 20°C . Sides of furnace are 1m each. Neglect heat loss due to radiation and convection from base. (9)
- VI. (a) A sphere of radius 5cm is concentric with another sphere. Find the radius of the outer sphere so that shape factor of the outer sphere with respect to the inner sphere is 0.6 (5)
- (b) A double walled flask may be idealized to be equivalent to two infinite parallel plates. The emissivities of walls are 0.3 and 0.7 respectively. The space between them is evacuated. A shield of polished aluminium of $E = 0.05$ is inserted between them. Find the reduction in heat transfer due to insertion of radiation shield. (10)

OR

- VII. An enclosure consists of a rectangular parallel piped 1m x 2m x 4m. One of the 1m x 2m surfaces act as a black surface at 475K and other acts as a black surface at 375K . If the other four surfaces of the enclosure act as re-radiative surfaces find the temperature of the re-radiative surfaces and the net radiative heat transfer between the two active surfaces. (15)
- VIII. Two identical counter flow type of heat exchangers are available water at the rate of 3600 kg/hr and at 30°C is to be heated by cooling an oil ($C_p = 2.1\text{ KJ/KgK}$) at 90°C . The oil flow rate is 2700 kg/hr . The heat exchangers are connected in series on the waterside and in parallel on the oil side. The oil flow rate is split in the ratio 2:1. Water enters the first heat exchanger at 30°C . Find the final water and oil temperatures. Overall heat transfer co-efficient in each heat exchanger is $300\text{ W/m}^2\text{K}$. Area available per heat exchanger is 4m^2 . (15)

OR

- IX. (a) Define LMTD and derive its expression for parallel flow heat exchangers. (10)
- (b) An existing heat exchanger of 20m^2 surface area is to be used to condense steam at 100°C . The cooling medium will be feed water available at 40°C , its flow rate being 0.9kg/s . The overall heat transfer coefficient is estimated to be $120\text{ W/m}^2\text{K}$. Calculate the exit temperature of feed water. (5)
